# JONATHAN MACÍAS

**Senior Software Engineer — .NET · Azure · AI Integration**

Cúcuta, Colombia · UTC-5 (full US Eastern/Central overlap) · Open to remote

jthmacias1995@gmail.com · +57 315 596 3394 · profile-page-eight.vercel.app

linkedin.com/in/jonathan-macias-mora · github.com/jonathan-macias

## SUMMARY

Software Engineer with 5+ years building backend systems, web platforms and integrations on .NET and Azure for international clients in financial services, fintech and retail. Strong in ASP.NET Core, React / Next.js, SQL Server and PostgreSQL, with production experience in microservices (RabbitMQ, CQRS), Python / FastAPI, a .NET Framework to .NET 8 cloud migration and Power Platform automation. I build LLM applications end to end: AI agents with tool calling, MCP servers, RAG on pgvector and eval suites gated in CI.

## SKILLS

**Backend:** C#, .NET 10 / .NET 8, ASP.NET Core Web API, .NET Framework 4.8, Entity Framework Core, LINQ, REST APIs, Microservices, CQRS, Python, FastAPI, Pydantic, SQLAlchemy / SQLModel

**AI / LLM:** AI agents, tool calling, structured outputs, RAG, pgvector, MCP (Model Context Protocol), LLM evals (LLM-as-judge), prompt-injection defense, Pydantic AI, Google Gemini, Claude Code

**Frontend:** React, Next.js, TypeScript, JavaScript

**Cloud & DevOps:** Azure (App Service, Storage, Key Vault, Azure SQL, Entra ID), AWS (Cognito), Docker, GitHub Actions, CI/CD, Git

**Data & Messaging:** SQL Server, Azure SQL, PostgreSQL, Dataverse, stored procedures, query optimization, RabbitMQ, Twilio (WhatsApp), Mailgun

**Power Platform:** Power Apps (Model-Driven, Canvas, PCF), Power Automate, Custom Connectors, Power BI

**Security & Quality:** Microsoft Entra ID, Amazon Cognito, OAuth 2.0 / OIDC, JWT, RBAC, xUnit, pytest, TDD, SOLID

## EXPERIENCE

**Nelumbo Consultores — .NET Developer (Azure & Power Platform)** **Jan 2025 – Present**

*Software consultancy · International clients in Mexico across financial services, leasing and fintech*

- Built an asset-remarketing platform (.NET 10, React, SQL Server) with an internal back office and a customer portal for auctions and direct sales: bidding, accept / reject / relaunch workflows, personalized dashboards and role-based access.
- Secured internal and customer users with Microsoft Entra ID and Amazon Cognito (JWT, RBAC); synced data from client data warehouses, integrated Mailgun and deployed on Azure App Service with Storage and Key Vault.
- Reduced SQL Server / Azure SQL query execution time by 35% by rewriting stored procedures and redesigning indexes.
- Developed .NET microservices for a prepaid-card payroll platform (worker onboarding, card assignment, deposits through external card issuers) using CQRS, RabbitMQ and PostgreSQL on AWS.
- Built a Python / FastAPI gateway that centralizes and secures access to data warehouse APIs in a restricted network, so consumer applications never reach internal data systems directly.
- Delivered Power Platform solutions for collections, lease terminations and WhatsApp campaigns (Model-Driven and Canvas apps, PCF, Dataverse, Twilio, Outlook 365, Power BI): rule-based overdue-account notifications, agent assignment, communication timelines and Excel-driven dynamic messaging, with Power Automate flows that eliminated 15+ hours per week of manual work.

**Epsilon Tech SAS (formerly CS Software SAS) — Full-Stack .NET Developer** **Aug 2021 – Jan 2025**

*Commercial, accounting and e-invoicing software for retailers and restaurants*

- Built the cloud version of the company's accounting and e-invoicing platform on .NET 8, ASP.NET Core and React / Next.js, migrating it from the .NET Framework 4.8 monolith that I also maintained.
- Containerized services with Docker and built CI/CD pipelines to Azure App Service, cutting deployment time by 40% with zero-downtime releases.
- Integrated electronic-invoicing providers and POS systems, keeping sales and accounting data consistent.
- Improved PostgreSQL response times by 30% through indexing and query optimization; mentored junior developers on unit testing and TDD.

*Earlier: IT Support Technician — Epsilon Tech SAS (formerly CS Software SAS) (2019 – 2021)*

## AI PROJECTS

**Support Triage Agent** — *Python, FastAPI, Pydantic AI, Gemini, MCP, PostgreSQL*
github.com/jonathan-macias/support-triage-agent

- AI agent that triages support tickets: it calls tools for customer history and similar tickets, then returns a validated priority, category, summary and reply in the customer's language, flagging mass incidents and repeat issues.
- One service layer shared by a FastAPI API (7 endpoints), the agent and an MCP server (6 least-privilege tools) for Claude Desktop and Claude Code.
- Eval-driven development: 11 live-model cases with a calibrated LLM judge (98% pass rate) and 36 pytest tests gated in GitHub Actions; layered prompt-injection defenses and a fallback model for resilience.

**MessageFlow** — *.NET 10, Next.js, PostgreSQL + pgvector, RAG, Gemini*

- Full-stack WhatsApp campaign platform: Excel datasets, dynamic templates and scheduled background flows through Evolution API, with a model-agnostic LLM layer (Gemini today) that drafts templates from natural language and pgvector-based RAG.
- ASP.NET Core Web API with CQRS and EF Core, ASP.NET Core Identity with Google OAuth, and a Next.js / React / TypeScript admin UI.

## EDUCATION

**B.S. in Systems Engineering** — Fundación de Educación Superior San José, Colombia (2023)
**Technologist in Information Systems Analysis and Development** — SENA, Colombia (2016)

## LANGUAGES

Spanish (native) · English (professional working proficiency)